PreLab 6

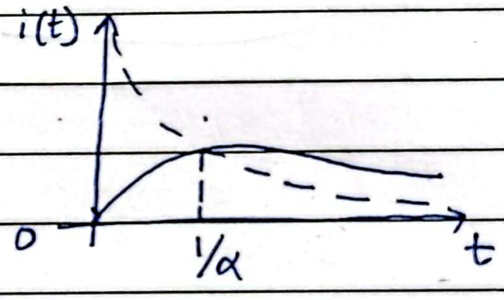
Date:

Q.1) For series and parallel RLC circuits the conditions for over-damped, under-damped and critically damped circuits are:

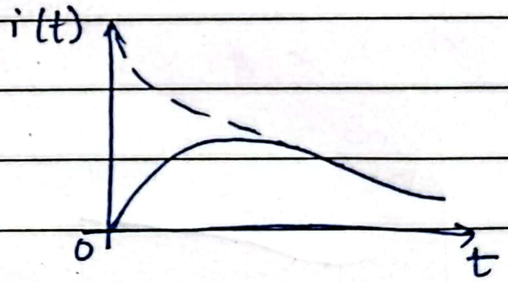
Overdamped: $R^2 > 4L/C$
Underdamped: $R^2 < 4L/C$
Critically damped: $R^2 = 4L/C$

Series RLC circuit

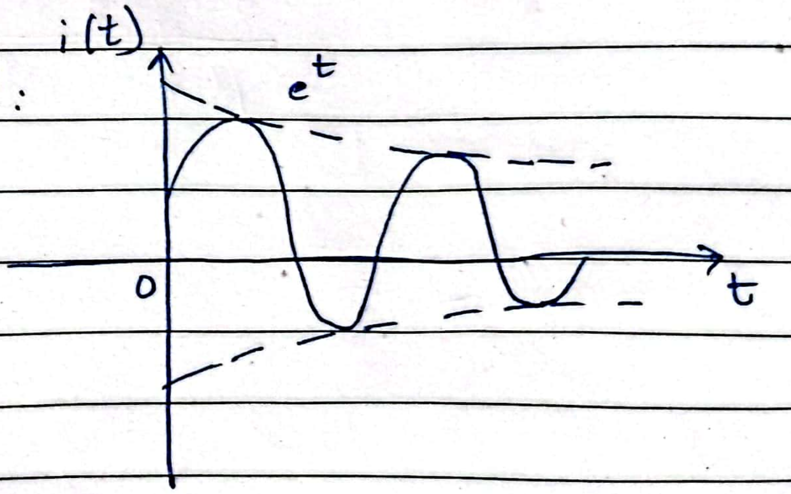
Critically damped:



Over damped:



Underdamped:



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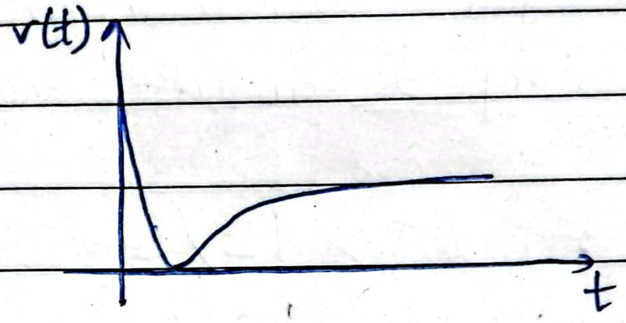
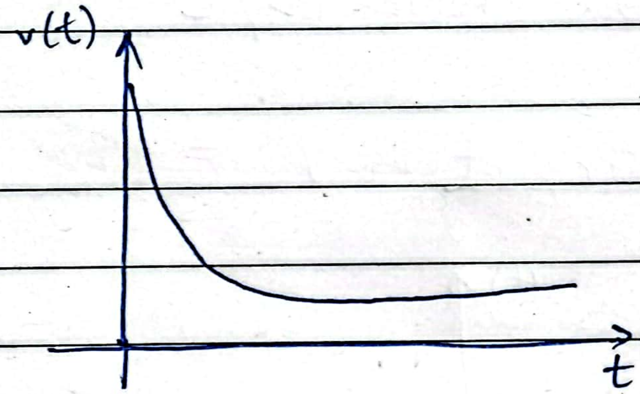
Parallel RLC Circuit:

Under ~~Critically~~ damped :

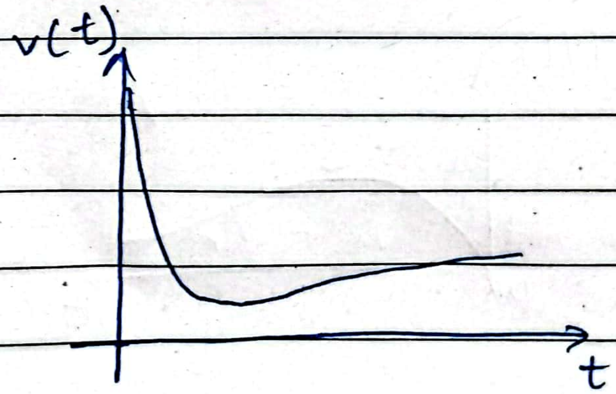


Overdamped :



~~Underdamped~~ :

Critically damped:



Q.2) $X_c = \dfrac{-j}{2\pi f C}$, $X_L = j2\pi f L$

$|z| = \sqrt{R^2 + (X_L - X_c)^2}$, $\phi = \tan^{-1}\left(\dfrac{X_L - X_c}{R}\right)$

$R = 1k\Omega$, $L = 10mH$, $C = 10nF$

At $f = 8kHz$:

$X_c = \dfrac{-j}{2\pi(8k)(10n)} = \boxed{1989.4\Omega}$, $X_L = j2\pi(8k)(10m) = \boxed{502.7\Omega}$

$\Rightarrow X_L - X_c = -1486.7$

$|z| = \sqrt{(1k)^2 + (-1486.7)^2} = \boxed{1791.8\ \Omega}$

$\phi = \tan^{-1}\left(\dfrac{-1486.7}{1000}\right) = \boxed{-56.07^\circ}$

At $f = 12kHz$:

$X_c = \dfrac{-j}{2\pi(12k)(10n)} = \boxed{1326.3\ \Omega}$, $X_L = j2\pi(12k)(10m) = \boxed{753.98\ \Omega}$

$X_L - X_c = -572.32$

$|z| = \sqrt{(1k)^2 + (-572.32)^2} = \boxed{1152.2\ \Omega}$

$\phi = \tan^{-1}\left(\dfrac{572.32}{1000}\right) = \boxed{-29.78^\circ}$

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At $f = 16k\ Hz$:

$X_c = \frac{-j}{2\pi(16k)(10n)} = \boxed{994.7\,\Omega}$, $X_L = 2\pi(16k)(10m) = \boxed{1005.3\,\Omega}$

$X_L - X_c = 10.6$

$|Z| = \sqrt{(1000)^2 + (10.6)^2} = \boxed{1000.06\,\Omega}$

$\phi = \tan^{-1}\left(\frac{10.6}{1000}\right) = \boxed{0.607°}$

At $f = 20k\ Hz$:

$X_c = \frac{-j}{2\pi(20k)(10n)} = \boxed{795.78\,\Omega}$, $X_L = 2\pi(20k)(10m) = \boxed{1256.6\,\Omega}$

$X_L - X_c = 460.86$

$|z| = \sqrt{(1k)^2 + (460.86)^2} = \boxed{1101.\cancel{9}52}$.

$\phi = \tan^{-1}\left(\frac{460.86}{1000}\right) = \boxed{24.74°}$.

At $f = 24kHz$:

$X_c = \frac{-j}{2\pi(24k)(10n)} = \boxed{663.15\,\Omega}$, $X_L = 2\pi(24k)(10m) = \boxed{1508\,\Omega}$

$X_L - X_c = 814.8\,\Omega$

$|Z| = \sqrt{(1k)^2 + (814.8)^2} = \boxed{1309.1\,\Omega}$

$\phi = \tan^{-1}\left(\frac{814.8}{1000}\right) = \boxed{39.2°}$.

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At $f = 30k\ Hz$:

$X_c = \frac{-j}{2\pi(30k)(10n)} = \boxed{530.5\ \Omega}$, $X_L = 2\pi(30k)(10m) = \boxed{1885\ \Omega}$

$X_L - X_c = 1354.46\ \Omega$

$|Z| = \sqrt{(1000)^2 + (1354.46)^2} = \boxed{1683.6\ \Omega}$

$\phi = \tan^{-1}\left(\frac{1354.46}{1000}\right) = \boxed{53.56^\circ}$

Freq (Hz)	X_C	X_L	$Z\angle\phi$
8k	1989.4 Ω	502.7 Ω	$1791.8\angle -56.07^\circ$
12k	1326.3 Ω	753.98 Ω	$1152.2\angle -29.78^\circ$
16k	994.7 Ω	1005.3 Ω	$1000.06\angle 0.607^\circ$
20k	795.78 Ω	1256.6 Ω	$1101.1\angle 24.74^\circ$
24k	663.15 Ω	1508 Ω	$1309.1\angle 39.2^\circ$
30k	530.5 Ω	1885 Ω	$1683.6\angle 53.56^\circ$